



FIRST SEMESTER 2026-27

Course Handout (Part II)

Date: 01/08/2026

In addition of Part 1 (General Handout for all courses appended to the Time Table), this portion gives further specific details regarding the course.

Course Number : EEE G515

Course Title : Real-Time Embedded Systems

Instructor-in-charge : G S SSHA CHALAPATHI

Course Description :

This course will introduce the general concepts related to real-time systems from introduction, to designing a real-time system (RTS). It will deal with a reference model of an RTS which can be applied to any RTS. It will also teach the scheduling algorithms, resource management techniques, issues in hardware and software design, Real-Time Operating System (RTOS), advanced concepts of software design and validation techniques for these systems.

1. Scope and Objectives:

Real-time systems are those systems that require a deterministic upper time-bound for the execution of a job. With the proliferation of the embedded system, which is the heart of items ranging from common every day appliances to life-critical systems, a great deal of work has gone into the characterization and design issues of Real-time systems. The course focuses on the basics of Hardware design, advanced concepts of software design and validation techniques for these systems. Specifically we discuss the Operating system, Scheduling, Software techniques, Run-time environment, failure management algorithms, Reliability and Fault-tolerance of real-time systems etc.

2. Text Book:

1. Liu Jane W. S.: Real-Time Systems, Pearson Education, India 2003.

3. Reference Books:

1. Krishna C.M. & Shin K.G.: Real-Time Systems, McGraw-Hill 1997
2. Laplante Phillip A.: Real-Time System Design and Analysis. Third Edition PHI 2005.
3. Qing Li, Caroline Yao. Real-Time concepts for Embedded systems CMP Books
4. Raj Kamal, Embedded Systems Architecture, programming and Design, Tata McGraw Hill
5. IEEE or related journal papers on some of the course topics.





4. Course Plan:

Module no	Lecture No	Topic	Reference	Learning Outcomes
1.	1-4	Introduction and Basic Real-Time Concepts	Ch 1-4 T1	To introduce basic concepts in a an RTS
2.	5-14	Real-Time Scheduling algorithms	Ch 5-7 T1	To learn Real-Time Scheduling algorithms
3.	15-22	Resource Access Protocols	Ch 8 T1	To learn Resource Management Protocols
4.	24-25	Real-Time Operating Systems	Ch 12 T1 , Ch 3 R2, Ch 9 R4	To understand the basics of RTOS
5.	26-27	Inter-task Communication and Synchronization	Ch 3 R2	To understand the Inter-task Communication and Synchronization in RTOS
6.	28-29	Memory Management and File systems	Ch 3 R2/Class Notes	To understand the Memory Management and File systems
7.	30-31	Scheduling algorithms for multi-processors	Class Notes	To understand scheduling policies in multi-processors
8.	32-34	Real-Time specification and Design Technique	Ch 4 R2	To understand the specification and Design Technique in RTS
9.	35-36	Hardware Issues	Ch 2 R2	To learn the hardware issues in RTS
10.	37-38	Software Issues	Ch 5 R2	To learn the software issues in RTS
11.	39-40	Fault tolerance and Reliability	Ch 8 R2	To learn the concepts related to Fault tolerance and Reliability

Total Lecture Hours

40





5. Evaluation Scheme

EC No.	Evaluation component	Duration	Weightage	Date, time and venue	Nature of component
1	Mid Semester Test	90 min	25 %	08-Oct-2026 AN2	Open Book/Closed Book
2	Assignments and classroom interaction	To be announced	40%	To Be announced	Open book
3	Comprehensive exam	3 hours	35 %	12-Dec-2026 AN	Open Book/Closed Book

6. **Chamber Consultation Hour** : To be announced by the respective campus instructor
7. **Make-up Policy**: Make Up will be given only in genuine case and with prior permission
8. Notices regarding this course will be displayed be sent by the respective campus instructor via BITS emails.

Instructor-in-Charge

EEE G515

